

CHAPTER-2

SHAPING OF THE EARTH'S SURFACE

1. What are the sources of energy required to cause movements associated with the internal forces of the Earth?

1. The main source of energy for internal movements is the intense heat present in the Earth's core.
2. This heat warms the material in the mantle and makes it move slowly.
3. The hotter and lighter mantle material rises, while the cooler and heavier material sinks.
4. This continuous circulation produces convection currents in the mantle.
5. These currents push and pull the tectonic plates, causing folding, faulting, earthquakes, volcanic eruptions and the formation of major landforms.

2. Relate various physiographic divisions with the endogenic forces responsible for their origin.

1. **The Himalayan Mountains:** They were formed by folding when the Indo-Australian Plate collided with the Eurasian Plate at a convergent boundary.
2. **The Northern Plains:** A large tectonic depression developed between the rising Himalayas and the Peninsular Plateau; it was later filled with river deposits.
3. **The Peninsular Plateau:** It is an ancient and stable part of the Earth's crust shaped by uplift, faulting and volcanic activity.
4. **The Coastal Plains:** They developed partly through the uplift and subsidence of coastal margins and were later modified by river and marine deposition.
5. **The Islands:** The Andaman and Nicobar Islands are connected with tectonic and volcanic activity at a plate boundary, while Lakshadweep Islands are mainly of coral origin.

3. Why and where do earthquakes occur frequently? Is it possible to predict earthquakes?

Why and where do earthquakes occur?

1. Earthquakes occur when stress created by moving tectonic plates is suddenly released along plate boundaries or faults.
2. They occur frequently at convergent and transform boundaries, especially around the Pacific Ocean in the **Ring of Fire**.

Is it possible to predict earthquakes?

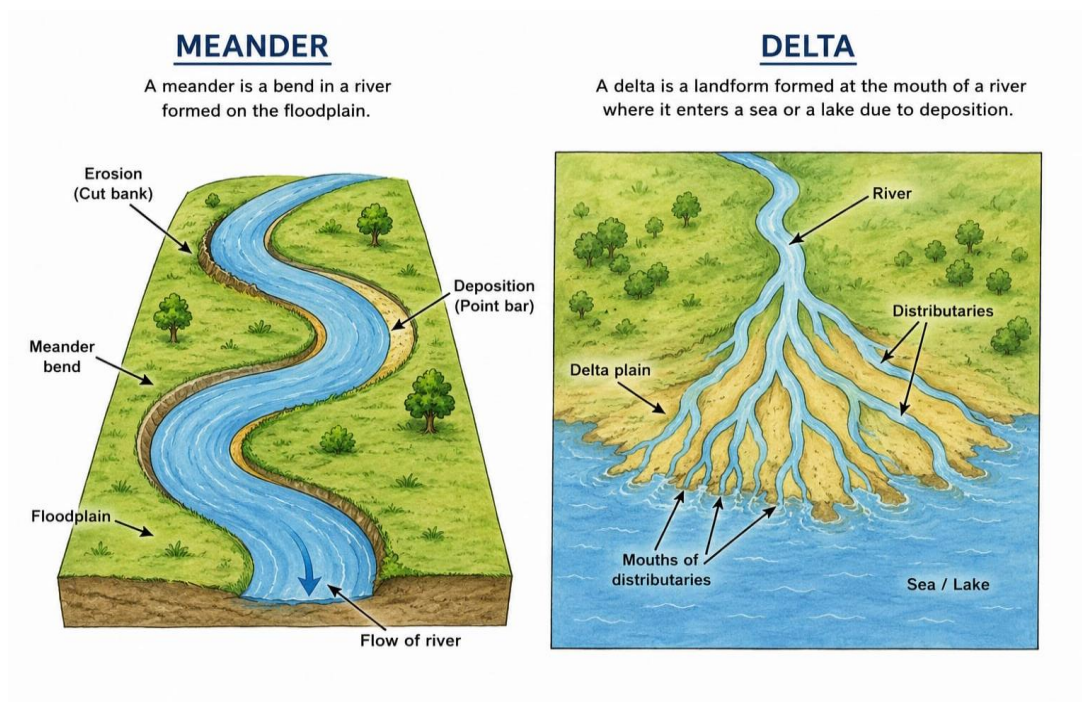
1. The exact time, place and magnitude of an earthquake cannot be predicted accurately.

2. Scientists can identify earthquake-prone regions by studying plate boundaries, faults, past earthquakes and seismic activity.
3. Such identification helps in disaster preparedness, safer construction and reducing the loss of life and property.

4. “Plate movements are responsible for the distribution of earthquakes and volcanoes.” Explain.

1. Tectonic plates move slowly over the semi-molten asthenosphere because of convection currents in the mantle.
2. At **convergent boundaries**, plates collide, and an oceanic plate may sink beneath another plate, causing earthquakes and volcanic eruptions.
3. At **divergent boundaries**, plates move apart and magma rises to form new crust, producing volcanic activity and earthquakes.
4. At **transform boundaries**, plates slide past one another, creating friction and frequent earthquakes.
5. Therefore, most earthquakes and volcanoes are concentrated along plate boundaries, particularly around the Pacific Ocean’s **Ring of Fire**.

Q5. Draw and label a diagram of a meander and a delta.



6. How are deforestation and erosion associated with each other? Explain.

Answer :-

1. Trees and plants protect the soil from the direct impact of rainfall and strong winds.
2. Their roots bind soil particles together and prevent the fertile topsoil from being carried away.
3. Deforestation removes this protective vegetation cover and leaves the land bare.
4. Rainwater then flows rapidly over the exposed surface, while wind easily blows away loose soil particles.
5. Therefore, deforestation increases soil erosion, reduces soil fertility and may also make slopes unstable, leading to landslides.

7. Develop a plan to protect the land in your local area from erosion.

Answer :-

1. Plant trees, grasses and shelter belts on open land and along roads, fields and water channels to hold the soil firmly.
2. Construct field bunds and contour trenches to slow surface run-off and allow rainwater to enter the soil.
3. Build small check dams across drains and streams to reduce the speed of water and encourage sediment deposition.
4. Prevent unnecessary tree cutting, overgrazing and unplanned construction, and organize regular community monitoring of erosion-prone places.

8. Which disasters might occur in your region? Give a mitigation plan.

Answer :- For the Hussainpur–Punjab region, the following disasters may occur:

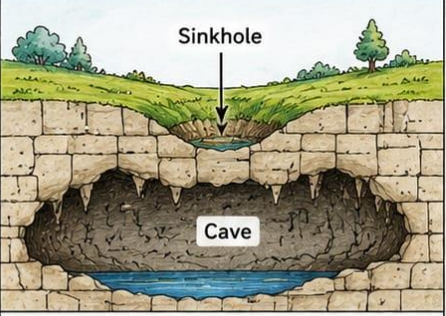
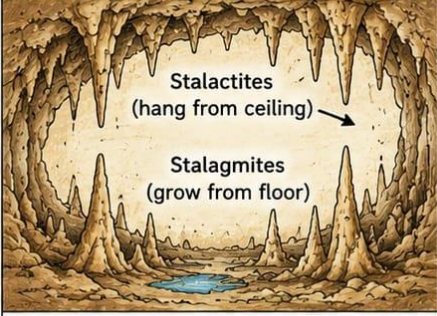
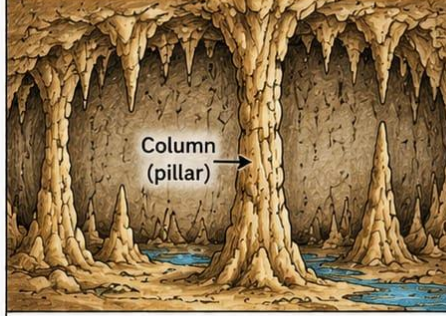
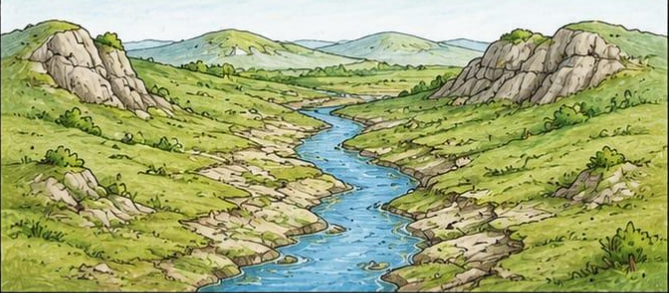
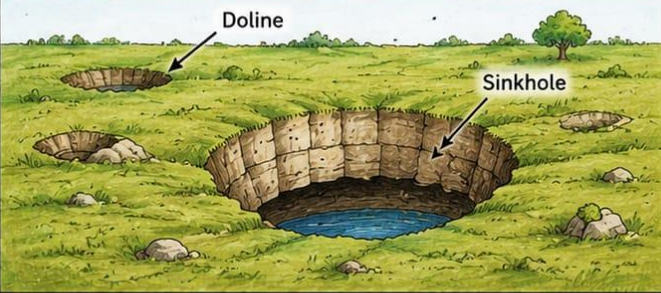
1. **Floods and waterlogging:** Drains should be cleaned before the monsoon, flood-prone places should be identified and safe evacuation routes should be marked.

2. **Earthquakes:** Buildings should follow earthquake-resistant designs, heavy furniture should be secured and students should practice “Drop, Cover and Hold.”
3. **Dust storms and strong winds:** People should remain indoors, close windows, protect their faces and plant shelter belts around fields and settlements.
4. **Classroom mitigation plan:** Prepare an emergency contact list, first-aid kit, assembly point and student disaster-management teams, and conduct regular evacuation drills.

9. Prepare a model of landforms created by underground water.

LANDFORMS CREATED BY UNDERGROUND WATER

Underground water dissolves soluble rocks like limestone and creates different landforms.

1. SOLUTION CAVES  Underground water dissolves limestone rocks and forms large underground openings called caves.	2. STALACTITES AND STALAGMITES  Water dripping from the roof of caves forms stalactites. Deposits on the floor build up to form stalagmites.	3. STALACTITE COLUMN (PILLAR)  When stalactites and stalagmites grow and join together, they form a column or pillar.
4. LIMESTONE VALLEYS  Underground water dissolves the limestone rocks and broad, flat-bottomed valleys are formed.	5. SINKHOLES AND DOLINES  When the roof of underground caves collapses, the land surface sinks and forms sinkholes or dolines.	

★ **Note:** These landforms are common in areas where limestone rocks are found, such as karst regions.

10. What precautionary measures will you take if you are staying in an earthquake-prone region?

1. I will ensure that the house is constructed according to earthquake-resistant building standards and avoid weak or damaged structures.
2. I will firmly secure cupboards, shelves, gas cylinders and other heavy objects so that they do not fall during an earthquake.
3. I will keep an emergency kit ready with drinking water, dry food, medicines, a torch, a first-aid box and important contact numbers.
4. During an earthquake, I will follow “Drop, Cover and Hold”, stay away from windows and lifts, and take shelter under a strong table.
5. After the shaking stops, I will switch off electricity and gas, move carefully to an open area, help injured persons and follow official safety instructions.

11. Prepare a map showing landform-associated disasters that happened in the current calendar year.

